Tandem Reductive Amination and Deuteration over a Phosphorus-modified Iron Center

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Introduction

Nitrogen-containing moieties existed extensively in pharmaceuticals, agrochemicals, biomolecules, and polymers.[1] So, developing methods for amines synthesis and valorization are of great importance. In our previous works, we for the first time reported a durable Ru_1/NC single-atom catalyst (SAC) for highly efficient reductive amination to produce primary amines (**Fig. 1a**)[1], and also an unique Ru_1Co_{NP} surface single-atom alloy catalyst (SSAAC) for new transformation of furfural amination towards N-heterocyclic piperidine synthesis (**Fig. 1b**)[2].

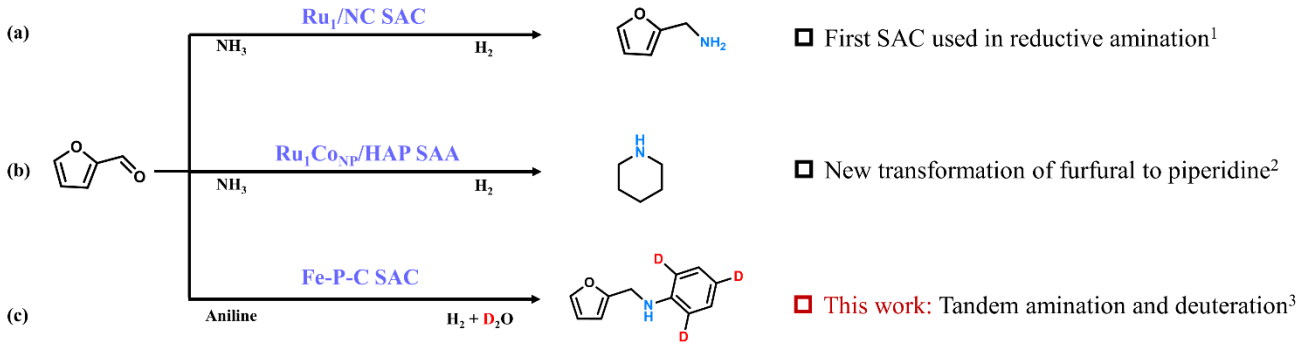

Figure 1. Previous developed (a) Ru_1/NC single-atom catalyst and (b) Ru_1Co_{NP} single-atom alloy catalyst for amines synthesis, and (c) this work of Fe-P-C single-atom catalyst for deuterated amines synthesis

Deuterated amines are key building blocks for drug synthesis and the identification of metabolites of new pharmaceuticals. In 2007, Concert Pharmaceuticals sold a deuterated drug, Kalydeco, used in treating cystic fibrosis, to Vertex Pharmaceuticals[4], illustrating the potential for deuterated pharmaceuticals. Thus, the quest for efficient and widely applicable methods for the selective synthesis of such deuterated compounds continues to attract significant attention from academic and industrial researchers all over the world. In this work, we for the first time describe an Fe-P-C catalyst with highly dispersed isolated metal sites that enables tandem reductive amination deuteration sequences towards synthesis of deuterated amines (**Fig. 1c**)[3].

Materials and Methods

For the preparation of Fe-P-C-800 catalyst, a mixture of 30 mg Fe $(NO_3)_3 \cdot 9H_2O$ and 4 g phytic acid solution was added to 50 mL H_2O and reflux at 120 °C, followed by addition of 2.0 g fumed silica. Then, the H_2O was removed, and the remaining solid was transferred to a crucible in the furnace and heated to 800 °C for 2 h in Argon atmosphere. The obtained black powder was washed by NaOH solution to remove the SiO_2 support to obtain Fe-P-C-800.

In the typical reaction, 0.25 mmol *p*-anisidine 1a, 0.25 mmol benzaldehyde 2a, 20 mg Fe-P-C-800 catalyst, 405 μL D_2O, 1.0 mL toluene were added into autoclave and purged with 40 bar H_2. Then the reaction was performed at 140 °C for 24 h.

Results and Discussion

We have successfully synthesized an optimal material uses a phosphorus-doped carbon support which, in combination with the Fe centers (**Fig. 2b**), enables to carry out catalytic one-pot tandem processes for reductive amination and deuteration reactions to directly produce deuterated anilines/amines (**Fig. 2a**). This optimal phosphorus-modified Fe-based catalyst shows excellent performance in terms of both reactivity and regioselectivity for a wide range of deuterated anilines, amines, bioactive complexes, and drugs (> 50 examples, **Fig. 2c**). Coupled with DFT calculation and control experiments, we have shown that the Fe-P/C pair sites exhibit moderate activation ability for D_2O, while the Fe sites facilitate H_2 activation, which working together enable the tandem catalysis to be effectively achieved.

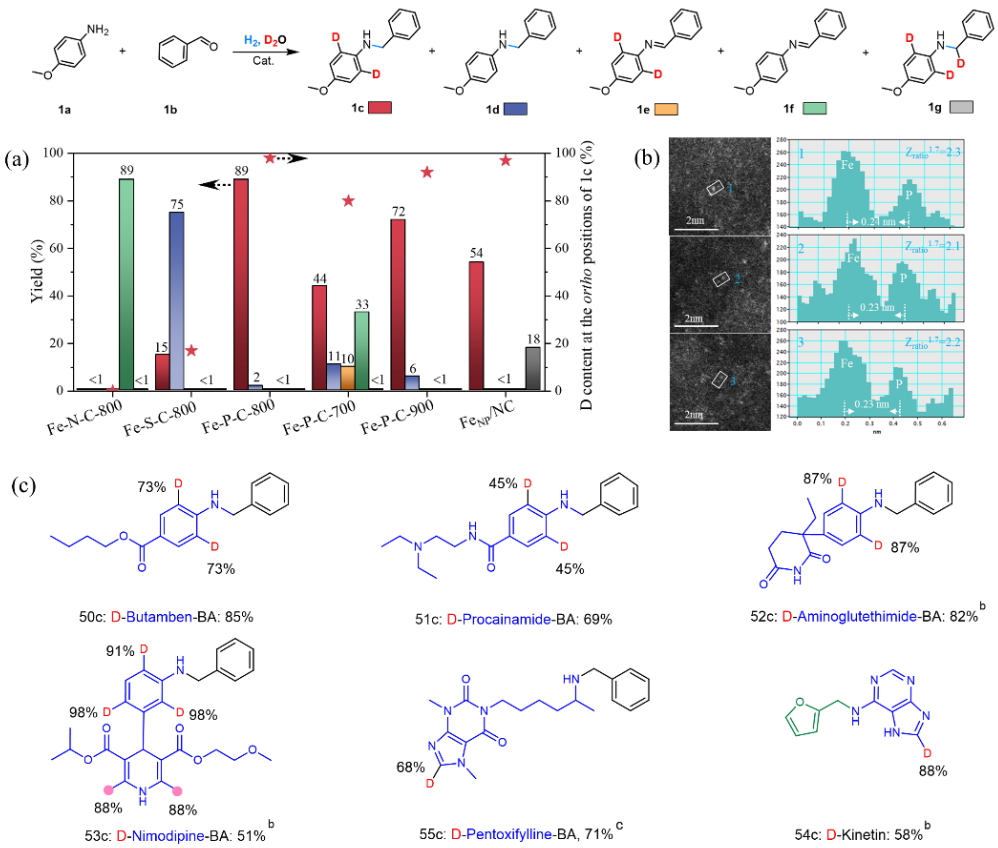

Fig. 2. (a) Catalytic performance; (b) HRTEM image; (c) Late-stage modified amination and deuteration of drugs.

Significance

This newly developed methodology may inspire and unfold a new perspective for design of multifunctional single-atom catalysts in other value-added organic synthesis.

References

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